

LAWRENCE PRIVATE COLLECTION

A Portal for Trophy Property — Public Collection + Members' Off-Market Section

Complete Technical Specification, Build Brief & Operating Manual — Version 2.0 (definitive)

0. How to read this document

This is the complete brief for **Lawrence Private Collection**: a portal listing exceptional property from roughly €20 million, built as a **normal, fully public, fully indexable listings site**, with one addition — an **Off-Market section that requires a free account**.

There is deliberately **no NDA, no proof of funds and no manual vetting in this phase**. A visitor creates an account, confirms their email, and the off-market listings open. That is the whole gate. Everything heavier (NDA signature, financial capability confirmation, per-member watermarking, manual approval) is designed for but **not built** — §23 specifies exactly how each is switched on later without a migration or a rewrite.

The same applies to the multi-agency backoffice: the data model carries the agency fields from day one, but Phase 1 runs as a single-team CMS. Turning on external agency accounts is a configuration and UI task, not an architectural one (§9.4).

It is written for one senior full-stack developer working with **Cursor + Claude**, and designed so the developer reaches **90-100 % of a launch-ready site without asking the client a single question**. Copy (§12), design values (§10.2), workflows (§8, §9), **where every piece of content comes from (§13)** and 48 product decisions (§21) are all pre-answered.

Normally asked	Answered in
"Where do I get photos, copy, market figures, geodata?"	§13 — every asset, named source, licence position
"What copy goes here?"	§12 — complete English microcopy
"What colours, fonts, spacing, logo?"	§10 and §13.4
"How does off-market access work?"	§8 — registration only, nothing more
"Which decisions are mine?"	§21 — 48 defaults, all reversible
"When do I stop and ask?"	§1.5 — four items, and none of them block building

Reading order: §1 → §2 → §6 → §8 → §13 → §21 → §22.

0.1 The two layers

Layer	Who sees it	Content
The Collection (public, indexable)	Everyone, including Google and AI assistants	Full listings with photography, facts, descriptions and maps. A normal, excellent luxury portal.
Off-Market (members, free account)	Anyone with a confirmed email address	Discreet listings the seller does not want published: full detail, but never indexed, never in a sitemap, never in a public API response.

The Collection is the acquisition engine. Off-Market is the reason people create an account — and the account is what turns anonymous traffic into a contactable audience. That is the entire business logic of this build.

0.2 Two delivery tracks

Track A — the fast launch (Prompts 1-14). Public Collection, Off-Market with simple registration, market-intelligence pages, sample inventory, SEO, performance, accessibility, deploy. Depends on nothing outside the repository.

Track B — optional, before or after go-live (Prompts 15-16). External agency accounts with roles, bulk import and a moderation queue. Every field it needs already exists from Prompt 3.

Build Track A end to end first. Do not begin Track B until the public site is live or demonstrably ready.

1. Developer operating method

1.1 /CLAUDE.md (create before any feature code)

```
# LAWRENCE PRIVATE COLLECTION – project rules for AI assistants

## What this is
A portal for trophy property from ~€20M. Two layers: a fully public, indexable Collection, and an Off-Market section that requires a free member account (email confirmation only – no NDA, no proof of funds, no manual vetting in this phase).
Next.js 15 App Router + Payload 3 + Postgres/PostGIS + Typesense, deployed on Vercel.

## Non-negotiable rules
1. Off-market content never leaks. It is never statically generated, never edge-cached, always noindex, never in a sitemap, OG image, RSS feed, public API response or 404 suggestion list, and always behind a server-side session check. Hiding it with CSS or a client-side conditional is a bug, not an implementation.
2. Every data-layer function takes an explicit Viewer argument. There is no ambient current-user global. A function that forgets the viewer must not compile.
3. Field visibility is decided by an ALLOWLIST projection per audience, never a denylist. A new model field is invisible everywhere until someone adds it to a projection deliberately.
4. Public performance budgets are acceptance criteria: LCP < 1.2s mobile, first-load JS < 110KB home / < 130KB listing gzip. Check `pnpm analyze` before committing.
5. No component or route queries the database directly; everything goes through /src/lib/db.
6. No hardcoded design values outside /src/tokens. No literal user-facing strings outside /src/messages.
7. Controlled enums over free text, always.
8. Sample listings always carry isSample=true, show a SAMPLE badge, emit noindex and are excluded from sitemaps.
9. WCAG 2.2 AA everywhere, including the account area.
10. Agency multi-tenancy fields exist on every relevant collection from day one and every query is written agency-aware, even though Phase 1 runs single-team. Never remove them "for simplicity".

## Definition of done for any task
pnpm typecheck && pnpm lint && pnpm test && pnpm test:e2e && pnpm analyze pass;
new logic has tests; `pnpm audit:exposure` passes; no console or hydration errors.

## audit:exposure
A required script that crawls every public route as an anonymous visitor and asserts that no response contains: an off-market listing id or slug, an exact address, owner data, internalValueEur, or a price for a listing whose priceDisclosure is not "exact". Must pass before every merge.

## Commands
pnpm dev · build · typecheck · lint · test · test:e2e · analyze · seed · sample:purge
search:reindex · audit:seo · audit:ally · audit:exposure · audit:all
```

1.2 `/.cursor/rules/lawrence.mdc`

The same rules, plus: *"Before implementing any feature that renders listing data, state which audience (anonymous, member, staff) sees which fields and where the server-side check happens. Then implement. Then self-check against the acceptance criteria and report pass/fail per line."*

1.3 `/DECISIONS.md`

Append-only. Date, decision, alternatives, reason, reversibility. This is what the client reviews at handover instead of being interrupted.

1.4 The build loop

Plan → implement on a branch → self-verify against the prompt's acceptance criteria → `pnpm audit:all` → commit and tag → log decisions. Small reversible commits. When the context degrades, start a fresh chat anchored on `CLAUDE.md` plus the relevant spec section.

1.5 The only four things that require the client

Everything else is settled by §21 or decided by the developer and logged.

1. Domain, logo approval and legal entity details for the footer and policies.
2. Legal review of the policy placeholders before go-live (§13.11) — does not block any development.
3. Go-live date and DNS access.
4. Staging sign-off.

2. Product definition

2.1 One-line definition

Lawrence Private Collection: the world's trophy homes, published openly — and the ones that are not, kept for members.

2.2 Admission policy (enforced in code)

Default threshold: €20 M or local-currency equivalent at the daily FX snapshot.

Tier	Range	Behaviour
prime	€10-20 M	Exception track, admin decision, capped at 10 % of published inventory — for markets where €20 M is effectively the ceiling (parts of Italy, Greece, Portugal, the Caribbean). Without this valve half of Europe looks empty.
trophy	€20-50 M	The core of the Collection
signature	€50 M+	Editorially prioritised, always featured
off_market	no floor, but in practice the same standard	Members only

Listings priced "on request" still carry an admin-only `internalValueEur` so the threshold can be enforced. That field is never exposed publicly under any circumstance.

2.3 Access model (deliberately minimal)

Audience	How	Sees
Visitor	—	The entire public Collection in full: photography, facts, description, locality-level map, price or "Price on request", agency contact, enquiry form. Plus market pages, intelligence and journal.
Member	Free account: email + password (or magic link), email confirmation, done	Everything above, plus the Off-Market section , saved listings with private notes, a requirements profile, and a "new off-market matches" email.
Staff	Invited	Everything, plus the CMS.

There is no application, no waiting list, no NDA, no document upload and no manual approval in Phase 1. Registration is two fields and a confirmation click. Friction is the enemy of the first 5,000 accounts; exclusivity in this phase is expressed by the *inventory*, not by the gate.

An `requireApproval` feature flag exists and is **off**. Turning it on later routes new accounts to a `pending` state with an admin queue — one flag, no migration (\$23.1).

2.4 Audiences

Audience	Need	Feature
UHNW buyers and their advisors	Quality inventory, discretion, no time wasted	Collection + Off-Market, requirements profile, direct desk contact
Prime brokers and agencies	International exposure, control over what is published	Per-listing publication control (\$8.4); Track B accounts later
Private sellers	Selling without the market knowing	Off-market placement, discreet owner submission form
Search engines and AI assistants	Authoritative market data	The Collection plus the market-intelligence library (\$15.4)

2.5 Scope of Phase 1 (Track A)

In: public Collection (home, browse, listing detail, market pages, intelligence, journal, editorial, contact), **member accounts with the Off-Market section**, saved listings and requirements profile, enquiry handling and routing, single-team CMS with full listing management, sample inventory of 45 listings, 6 languages, complete SEO + AI-discovery layer, accessibility to WCAG 2.2 AA, analytics, performance pass, staging and launch.

Track B (optional, \$9.4): external agency accounts, roles, bulk CSV import, moderation queue, agency dashboard.

Explicitly deferred with seams in place (\$23): NDA e-signature, financial capability confirmation, per-member forensic watermarking, manual member vetting, named per-listing access grants, in-app messaging, data rooms, payments, CRM, AI semantic matching, native apps.

3. Benchmarks

3.0 Legal boundary

We replicate interaction patterns and information architecture — conventions, not property. We never copy HTML/CSS/JS, scraped markup, photography, logos, licensed fonts, listing copy, editorial text or a competitor's inventory. Scraping a competitor's database is a terms breach and a legal risk, and is unnecessary (\$13).

3.1 Primary references

Christie's International Real Estate — <https://www.christiesrealestate.com> — the closest tonal target: restraint, serif typography, editorial pacing, auction-house authority. **Take:** type hierarchy, whitespace, the calm of the listing page, the `lifestyles/` URL structure as a landing-page model. **Avoid:** heavy hero video and a long scroll before the first property.

Sotheby's International Realty — <https://www.sothebysrealty.com> — a global network presented as a single brand, with country and region navigation at scale. **Take:** destination architecture, agent trust signals, the collection-page rhythm.

JamesEdition — https://www.jamesedition.com/real_estate — the scale benchmark (200,000+ listings from 16,000 luxury offices across 50+ countries) and proof that lifestyle-facet category pages drive organic acquisition. **Take:** the category-as-landing-page engine, applied here to market × segment combinations; the dense-but-calm card grid. **Avoid:** its marketplace feel — our positioning is the opposite.

Knight Frank — <https://www.knightfrank.com> and The Wealth Report — **the most important reference in this document**, not for listings but for research. Their published prime-market indices are why advisors know the name. §15.4 adapts that model directly: sourced, dated, citable data that makes us the source rather than a vendor.

Savills Research — <https://www.savills.com/research/> — the same lesson in repeatable formats that rank for years.

Vladi Private Islands — <https://www.vladi-private-islands.de/en/> — over 2,400 certified island sales and a masterclass in niche-authority tone from what is essentially a private register with a website. **Avoid:** dated execution — that gap is the opportunity.

Concierge Auctions — <https://www.conciergeauctions.com> — study their registration copy: qualification presented as part of the exclusivity rather than as friction. Our account creation should feel the same, at a fraction of the effort.

Barnes International — <https://www.barnes-international.com> · **Engel & Völkers** — <https://www.engelvoelkers.com> — useful for multi-locale structure and for how market pages are organised across countries.

3.2 Pattern references

Site	Take exactly this
airbnb.com	Map + list mechanics, in-card gallery, filter pills, saved-listing UX
compass.com	Listing-page rhythm: gallery → facts → story → map → agent
1stdibs.com	High-value item pages with restrained luxury e-commerce patterns
sothebys.com (auction house)	Provenance and lot storytelling — how a €60 M estate should be narrated
stripe.com · linear.app	Perceived performance, typographic discipline, instant transitions
mercury.com · wise.com	Account creation that feels premium rather than bureaucratic
rightmove.co.uk	Information density and speed on mobile

3.3 The thesis

At this price level the inventory is fragmented across brokerage websites that are slow, badly structured and invisible to search. Nobody has built a fast, editorially serious, data-rich portal at the top of the market with a genuine research library in front of it. The moat is twofold: the **research content**, which compounds and is what assistants cite, and the **member base**, which is an owned audience rather than rented traffic.

4. Design decisions (settled — do not re-open)

- Silence over spectacle.** Fewer elements per screen than any competitor. One idea per viewport. No badges, no ribbons, no "hot property", no countdown, no urgency devices.
- Monochrome plus one warm metal.** Obsidian, bone, graphite, with a muted aged-brass accent at $\leq 2\%$ of surface area (§10.2). No gold gradients, no black-and-gold cliché.
- Editorial typography.** High-contrast serif display, quiet grotesque body, letterspaced small-caps labels, **tabular figures for every price and measurement**.
- Photography full-bleed**, one uniform colour treatment, generous margins, never more than three images competing for attention.
- The Off-Market invitation is a whisper, not a wall.** On public pages it appears as a single quiet line with a count ("41 properties are held off-market"), never as a blur, a lock icon or an interstitial.
- Account creation is two fields.** Email and password, or a magic link. Nothing else on that screen.
- Slower motion.** 300–420 ms, no parallax, no scroll-jacking, `prefers-reduced-motion` honoured.
- Locality-level maps on public listings**, exact pins only where the seller permits (`coordinatePrecision`); off-market listings always show a circle, never a pin.
- No live chat widget, ever.** The desk is an enquiry form and a scheduled call.
- Empty states are computed**, never generic: name the most restrictive filter and offer one click to relax it.

5. Information architecture

5.1 Locales

6 locales: `en` (default), `it`, `fr`, `de`, `es`, `ru`. Adding one is a config entry plus a message file. Locale prefix always present; `/` performs an uncached 302 language detect. No trailing slash.

5.2 Public routes (indexable)

```
[/[locale]]                Home
[/[locale]/collection]     Browse the public Collection (canonical paths, $15.3)
[/[locale]/property/[slug]] Listing detail
[/[locale]/markets]        Market index (the research hub)
[/[locale]/markets/[market]] Market page: data, editorial, listings
[/[locale]/markets/[market]/[segment]] e.g. /markets/cote-dazur/waterfront-estates
[/[locale]/intelligence · /intelligence/[slug]] Reports (summary open, PDF requires an account)
[/[locale]/journal · /journal/[slug]] Editorial
[/[locale]/about · /discretion · /contact]
[/[locale]/sell]           Owner and broker submission
[/[locale]/join]           Create a free account (off-market access)
[/[locale]/legal/[slug]]  Privacy, cookies, terms, disclaimer
[/[locale]/404 · /410]     Designed error states
```

5.3 Member routes (all `noindex`, dynamic, never edge-cached)

```
[/[locale]/off-market]     The Off-Market section
[/[locale]/off-market/[id]] Off-market listing detail
[/[locale]/account]        Profile and preferences
[/[locale]/account/saved]   Saved listings with private notes
[/[locale]/account/requirements] Requirements profile (what they are looking for)
/api/member/*              Session-guarded endpoints
```

5.4 Backoffice and API

```
/admin/*                  Payload CMS (staff; Track B adds agency roles)
/admin/members             Member list, activity, export
/admin/enquiries           Enquiry queue with routing and status
/api/health · /api/revalidate · /api/og/[type]/[id]
/api/property/[slug]/brochure.pdf
/api/enquiry              Rate-limited intake
/sitemap.xml + /sitemaps/*.xml · /robots.txt · /llms.txt
/api/public/markets/[market]/stats Machine-readable market data ($15.6)
```

5.5 Market page grammar (the SEO engine)

Canonical: `/[locale]/markets/[market]` and `/[locale]/markets/[market]/[segment]`, where `segment` belongs to a controlled taxonomy: `waterfront-estates · vineyard-estates · ski-chalets · penthouses · private-islands · historic-estates · equestrian-estates · new-developments · golf-estates`.

Rules: one canonical URL per combination; a resolver normalises aliases with 301s; slugs come from the `Taxonomy` collection, localised per locale; a page renders only when it has published editorial copy plus at least three sourced data points. Empty combinations are never auto-published — that is the fastest route to a Helpful-Content demotion.

6. Data model

Payload 3 collections generate the TypeScript types used everywhere. Postgres 16 + PostGIS.

6.1 `Property` — identity and lifecycle

Field	Type	Notes
id	uuid	Off-market listings are addressed by id, never by a guessable slug
slug	text unique nullable	Public listings only; null for off-market
reference	text	Unique per agency
agency / agent	relations, agency required indexed	Present from day one , even though Phase 1 is single-team (\$9.4)
channel	enum required	public · off_market — the single routing decision
valueTier	enum	prime · trophy · signature
status	enum	draft · available · reserved · under_offer · sold · withdrawn · expired · archived
moderation	enum	unreviewed · approved · rejected · changes_requested + moderationNote — dormant in Phase 1, used by Track B
isSample	bool default false	Demo flag
featured	bool	Editorial only
publishedAt · expiresAt · lastVerifiedAt	timestamps	120-day freshness cycle (\$9.3)
fingerprint · duplicateOf	text indexed / relation	Duplicate detection
sourceType	enum	manual · csv_import · feed · owner_submission
viewCount · memberViewCount · enquiryCount	int	Off-market views counted separately

6.2 Commercial and structure

priceType enum fixed · on_request · price_band · auction · priceAmount numeric nullable · priceBandMin/priceBandMax numeric · currency enum EUR · USD · GBP · CHF · AED · SGD · HKD · priceEur generated indexed · **internalValueEur** numeric **required, admin-only, never serialised publicly** · priceDisclosure enum exact · band · on_request · tenure enum freehold · leasehold · usufruct · concession · ownershipStructure enum direct · spv · trust · foundation · company · saleStructure enum asset_sale · share_transfer · fractional · auction · annualRunningCostEur numeric nullable · taxNotes localised rich text (factual, never advice) · mandateType enum exclusive · open · introduction_only · commissionTerms text **staff-only** · availableFrom date.

6.3 Physical and provenance

propertyType enum: villa · estate · penthouse · townhouse · chalet · castle · palazzo · private_island · vineyard_estate · equestrian_estate · hotel_resort · development_site · apartment · lodge · ranch.

bedrooms · bathrooms · receptionRooms · builtAreaSqm · plotAreaSqm · plotAreaHa · terraceAreaSqm · floors · yearBuilt · renovatedYear · architect · heritageStatus enum none · listed · protected · unesco_area · provenance localised rich text · condition enum · staffAccommodation int · parkingSpaces int · energyRating.

features multi-enum: pool · indoor_pool · spa · gym · cinema · wine_cellar · ballroom · library · chapel · helipad · tennis · padel · golf_hole · equestrian · vineyard · olive_grove · beach · private_dock · marina_berth · ski_in_ski_out · gatehouse · staff_quarters · guest_houses · generator · geothermal · solar · smart_home · car_gallery · panic_room.

Optional waterfront sub-block (valuable at this level, and the bridge to the sister portal): waterAccess bool · waterBodyType enum · waterFrontageM numeric · mooringType enum · maxBoatLoaM numeric · berthCount int.

6.4 Location with disclosure control

location group: label (audience-aware display string) · addressLine **staff-only, never serialised to any audience** · locality · province · region · country ISO-2 indexed · continent · market relation → Market · coordinates PostGIS geography(Point,4326) GIST-indexed · coordinatePrecision enum exact · approximate_500m · locality_only (default approximate_500m) · publicGeography enum country · region · market · locality — the coarsest truthful label shown to anonymous visitors.

Implementation rule: the serialiser uses an **allowlist** of fields per audience, never a denylist. A new model field is invisible until someone deliberately adds it. This one decision prevents most exposure incidents.

6.5 Media and documents

- **Media** — variants at upload (320...3200 px, AVIF + WebP), **EXIF stripped without exception** (listing photos routinely carry the property's GPS), blurbhash LQIP, visibility enum public · members per asset (a listing can show 10 public images and hold 30 for members), minimum 2000 px long edge, credit/licence/sourceUrl, agency scoping. Public assets go to the public bucket; member-only assets go to the private bucket and are served through a signed URL route.
- **Document** — floor plans, brochures, inventories, surveys. Members-only by default, delivered via /api/secure/document/[token] with a short-lived signed URL. No watermarking in Phase 1 (the seam is \$23.3).

6.6 Members and engagement

Member (authenticated end user; separate from **User**, which is staff): `name` · `email` unique · `phone` optional · `country` optional · `preferredLocale` · `memberType` enum `buyer` · `advisor` · `broker` · `developer` · `other` (self-declared, optional, never gating) · `status` enum `active` · `pending` · `suspended` (default `active`; `pending` only when the `requireApproval` flag is on) · `emailVerifiedAt` · `marketingConsent` bool + `consentedAt` + `consentIp` · `lastActiveAt` · `source` enum `organic` · `off_market_cta` · `report_download` · `enquiry` · `referral` · `utm` json · `twoFactorEnabled` bool (optional for members in Phase 1) · **reserved and unused:** `ndaStatus`, `ndaSignedAt`, `capabilityStatus`, `capabilityBand`, `tier` — created now, never written to, so §23 is a flag rather than a migration.

SavedListing — member, property, note (private), `savedAt`.

Requirement — member · `budgetMinEur` / `budgetMaxEur` · `currency` · `markets` · `property types` · `must-have features` · `timeline` enum `immediate` · `6_months` · `12_months` · `opportunistic` · `notes` · `status` enum `active` · `paused` · `closed` · `notifyByEmail` bool. Used for the "new off-market matches your requirements" email and for the admin matching view.

MemberActivity — append-only: member, property, action enum `off_market_list` · `off_market_view` · `document_download` · `saved` · `enquiry` · `at` · `ip` · `userAgent`. Lightweight, but it is what lets the desk say to a seller "eleven members viewed your property this week", and it is the foundation of everything in §23.

Enquiry — name, email, phone, message, `property`, `agency`, `member` (when authenticated), `locale`, `source` enum `listing` · `off_market` · `market_page` · `report` · `contact` · `sell` · `desk_call`, `marketingConsent` + timestamp + IP, `utm` json, `status` enum `new` · `sent` · `viewed` · `qualified` · `spam`.

6.7 Content collections

Market — name (localised), slug per locale, country, region, polygon, centroid, hero image, localised intro, and a `stats` group maintained by editors: `medianPriceEurPerSqM`, `primeEntryEur`, `yoyChangePct`, `avgDaysOnMarket`, `transactionVolumeBand`, each with **source (URL)** and **asOfDate**. Every published number carries a source and a date — that discipline is what makes the research citable rather than decorative.

Report — title, slug, **ungated summary (400-600 words with the key findings and chart data)**, gated PDF (requires an account), charts JSON, publication date, authors.

Article, **Page** (block-based), **Taxonomy**, **Agency**, **Agent**, **Redirect**, **User** (staff roles `admin` · `editor`, plus Track B's `agency_admin` · `agency_agent`), **AuditLog** (staff actions), **ImportJob** (Track B), **FxSnapshot**.

6.8 Indexes

`slug` · `agency` · `channel` · `status` · `valueTier` · `priceEur` · `propertyType` · `country` · `market` · `expiresAt` · `fingerprint` · `isSample` · GIST on `coordinates` · `composite` (`channel`, `status`, `isSample`) for the hot public query · `Member(email)`, `Member(status)` · `MemberActivity(member, at)` and `(property, at)` · `SavedListing(member)`.

7. Stack and setup

7.1 Decisions

Area	Choice	Why
Framework	Next.js 15 App Router, TS strict	Collection static (SSG/ISR), Off-Market dynamic
CMS	Payload 3 embedded in the same app	No second server, Postgres-native, generated types, real admin UI, and access control granular enough for Track B later
DB	Postgres 16 + PostGIS (Neon)	Geo queries DB-side; PostGIS from day one
Search	Typesense Cloud , two collections: <code>public_listings</code> and <code>member_listings</code> , with separate keys — the browser only ever receives the public search-only key	Physical separation beats a filter flag
Member auth	Payload auth on the Member collection : email + password and magic link, email confirmation required, optional TOTP 2FA	No social login: it leaks the fact of membership to a third party
Styling	Tailwind v4 + token CSS variables	One file restyles everything
Maps	MapLibre GL + MapTiler (Protomaps if tile cost grows)	Locality circles and polygons; pins only where permitted
Media	Cloudflare R2 (public bucket + private bucket) + Cloudflare Images	Variants at upload, global CDN, signed URLs for member assets
i18n	next-intl , 6 locales	
Forms	Zod + React Hook Form	Shared schemas
Email	Resend + React Email	Enquiries, account, digests
Rate limit / sessions	Upstash Redis	Also signed-URL nonces
Bot protection	Cloudflare Turnstile on join, login, enquiry	Invisible, privacy-preserving
Scheduling	Cal.com embed for desk calls	
PDF	@react-pdf/renderer	Brochures
Analytics	Plausible on the public layer; member behaviour measured through <code>MemberActivity</code>	Privacy-first, no third party inside the member area
Monitoring	Sentry (PII scrubbed) + Vercel Speed Insights	
Testing	Vitest + Playwright + axe-core + Lighthouse CI	
Deploy	Vercel	ISR, previews, edge cache

7.2 Bootstrap

```
pnpm create payload-app@latest lawrence --template website --db postgres --use-pnpm
cd lawrence

pnpm add next-intl zod react-hook-form @hookform/resolvers maplibre-gl typesense \
  @react-pdf/renderer sharp blurhash resend react-email @react-email/components \
  @upstash/redis @upstash/ratelimit otplib qrcode papaparse date-fns clsx \
  tailwind-merge lucide-react @vercel/og i18n-iso-countries

pnpm add -D tailwindcss @tailwindcss/postcss autoprefixer vitest @vitejs/plugin-react \
  playwright @playwright/test @axe-core/playwright eslint prettier \
  prettier-plugin-tailwindcss typescript @faker-js/faker tsx dotenv-cli \
  @next/bundle-analyzer @lhci/cli

docker run -d --name lw-pg -p 5432:5432 -e POSTGRES_PASSWORD=lawrence \
  -e POSTGRES_DB=lawrence postgres/postgis:16-3.4
docker run -d --name lw-ts -p 8108:8108 -v /tmp/ts:/data typesense/typesense:27.1 \
  --data-dir /data --api-key=devkey --enable-cors

psql postgresql://postgres:lawrence@localhost:5432/lawrence \
  -c "CREATE EXTENSION IF NOT EXISTS postgis; CREATE EXTENSION IF NOT EXISTS pg_trgm;"

pnpm payload generate:types && pnpm dev
```

7.3 Environment variables

```

DATABASE_URL=                                PAYLOAD_SECRET=
NEXT_PUBLIC_SITE_URL=                        NEXT_PUBLIC_DEFAULT_LOCALE=en
R2_ACCOUNT_ID= R2_ACCESS_KEY_ID= R2_SECRET_ACCESS_KEY=
R2_PUBLIC_BUCKET= R2_PRIVATE_BUCKET=
NEXT_PUBLIC_CF_IMAGES_URL=                  NEXT_PUBLIC_MAPTILER_KEY=
TYPESENSE_HOST= TYPESENSE_ADMIN_KEY=
TYPESENSE_PUBLIC_SEARCH_KEY=                TYPESENSE_MEMBER_SEARCH_KEY=
UPSTASH_REDIS_REST_URL= UPSTASH_REDIS_REST_TOKEN=
SIGNED_URL_SECRET=
RESEND_API_KEY= DESK_NOTIFY_TO= NOTIFY_FROM=
TURNSTILE_SITE_KEY= TURNSTILE_SECRET_KEY=
UNSPLASH_ACCESS_KEY= PEXELS_API_KEY=
FX_API_URL=https://api.frankfurter.app
REVALIDATE_SECRET= CRON_SECRET=
NEXT_PUBLIC_PLAUSIBLE_DOMAIN=              SENTRY_DSN=
SAMPLE_DATA_ENABLED=true                   MEMBER_REQUIRE_APPROVAL=false

```

7.4 Folder structure

```

/src
/app/[locale]/(public)/...    home, collection, property, markets, intelligence, journal, join
/app/[locale]/(member)/...   off-market, account – force-dynamic, noindex, private cache
/app/(payload)/...           admin
/app/api/{health,og,revalidate,secure,enquiry,member,cron,public}
/components/{ui,property,collection,member,account,admin,layout}
/lib/db                      typed access; EVERY function takes a Viewer argument
/lib/access                  viewer resolution, audience projections
/lib/media                   signed URLs, EXIF stripping, variants
/lib/activity                MemberActivity helpers
/lib/search /lib/geo /lib/seo /lib/pdf /lib/email /lib/schemas /lib/match
/lib/services                EMPTY – future CRM, e-signature, KYC seams
/payload/{collections,access,hooks}
/tokens /messages /content /scripts /data

```

8. Members and the Off-Market section

8.1 The Viewer

Every data-layer call receives an explicit viewer. There is no ambient current-user global — a function that forgets it will not compile.

```

type Viewer =
| { kind: 'anonymous' }
| { kind: 'member'; id: string; status: 'active'|'pending'|'suspended' }
| { kind: 'staff'; id: string; role: 'admin'|'editor' }
| { kind: 'agency'; id: string; agencyId: string }; // Track B

```

8.2 Visibility rules

```

canSee(viewer, property) =
  (property.channel = 'public')
  ? status is publicly visible AND (isSample allowed by config)
  : viewer.kind = 'member' AND viewer.status = 'active'
OR viewer.kind = 'staff'
OR (viewer.kind = 'agency' AND property.agency = viewer.agencyId)

```

Off-market listings are excluded from: `generateStaticParams`, sitemaps, the public Typesense collection, OG image routes, RSS, related-listing suggestions, 404 recommendations, the public stats API, and every public JSON response. All of this is asserted by `audit:exposure` in CI.

8.3 Field projections (allowlist)

`/src/lib/access/projections.ts` — one table, exhaustively tested.

Field group	Anonymous (public listing)	Member (public listing)	Member (off-market)	Staff
Title, type, full facts	✓	✓	✓	✓
Full gallery (public assets)	✓	✓	✓	✓
Member-only assets	—	✓	✓	✓
Price	per priceDisclosure	per priceDisclosure	exact or band	exact
Location	publicGeography label + circle	locality + circle	locality + circle	exact
Exact coordinates / addressLine	never	never	never	✓
Description, provenance	✓	✓	✓	✓
Documents / floor plans	—	✓	✓	✓
Running costs, ownership structure	—	✓	✓	✓
internalValueEur, commissionTerms, owner data	never	never	never	✓

Note the shape of it: **the public Collection is genuinely open** — this is a normal portal, not a teaser. The account buys the off-market inventory, the extra imagery, the documents and the saved-listing tools. That is an honest value exchange, and honest value exchanges convert better than artificial scarcity.

8.4 Publication control on a listing

One required control in the CMS: "How should this property be published?"

Choice	Sets
Published openly	channel = public, price per the seller's instruction
Published without a price	channel = public, priceDisclosure = on_request
Published as a band	channel = public, priceDisclosure = band
Off-market (members only)	channel = off_market

Beside it, a live preview showing exactly what an anonymous visitor and a member would see. Brokers trust what they can be shown.

8.5 The registration flow (deliberately two minutes)

1. **Join** (/join) — email + password, or magic link. Optional: name, phone, country, what they are looking for. Marketing consent checkbox, unticked, optional. Turnstile. Nothing else.
2. **Confirm email** — one click. Account becomes active and the Off-Market section opens immediately.
3. (Optional, flag-gated and off by default) MEMBER_REQUIRE_APPROVAL=true sends new accounts to pending with an admin queue and a "we will confirm within one business day" message.
4. **First login experience** — a short, quiet welcome panel: here is the Off-Market section, here is how to save a listing, here is how to tell us what you are looking for.
5. **Requirements profile** (optional, prompted after the third off-market view) — budget, markets, types, timeline. Feeds the match email and the admin view.
6. Members may delete their account themselves; deletion anonymises MemberActivity and removes saved listings.

Copy throughout is in §12.3, and it never says "sign up" — it says "create an account to see the off-market collection".

8.6 Member media and documents

Member-only images and all documents live in the private bucket and are served through /api/secure/media/[token] and /api/secure/document/[token]: an HMAC over (assetId, memberId, expiry, nonce), **15-minute lifetime**, nonce burned in Redis, session re-checked server-side, Cache-Control: private, no-store, X-Robots-Tag: noindex, noimageindex. An unauthenticated request returns **404, not 403** — never confirm that an asset exists.

No watermarking in Phase 1; the pipeline is specified in §23.3 and can be enabled per asset later.

8.7 Activity and what it is for

MemberActivity is written on off-market list views, detail views, document downloads, saves and enquiries. It powers three things that matter commercially: the admin's view of demand per listing, the "eleven members viewed your property this week" line the desk gives sellers, and the match notifications. It is not analytics for its own sake, and no third-party script ever sees it.

8.8 Matching (small, high-value)

`/src/lib/match.ts` scores active `Requirement` records against new listings (budget overlap, market, type, must-haves). When a new off-market listing is published, matching members with `notifyByEmail` receive one discreet email. Admins see a matching board. One page and one function — deliberately isolated so it can become embedding-based later without touching anything else.

9. Backoffice

9.1 Phase 1: single-team CMS

Payload admin with staff roles `admin` and `editor`. Listing form grouped as: Essentials → **Publication** (the §8.4 control) → Commercial → Physical & provenance → Location → Media (with per-asset visibility) → Documents → SEO → Publishing. Live preview, drafts, versions, autosave, localised editing across 6 locales, media library with drag-and-drop gallery ordering, and a completeness meter.

9.2 Admin dashboard

Inventory by channel, market, value tier and status · members over time with source breakdown and the conversion from visitor to account · off-market views and document downloads per listing · enquiries per week by source with response status · requirements board with matches · a data-quality panel (missing images, short descriptions, missing market, expiring soon) · a warning if any sample content is publicly reachable. All queries via `/src/lib/db`, `Recharts`, read-only.

9.3 Listing freshness

120-day cycle. At T-14 the responsible editor gets a "confirm these are still available" list with one-click confirmation. On expiry: removed from search and sitemap, public URL returns **410 Gone** with a designed page suggesting the market page. Sold listings keep a "sold" state for 90 days without a price, then `noindex`, then 301 to the market page. Stale inventory is the fastest way to lose credibility at this level.

9.4 Track B: external agency accounts (optional, later)

Everything needed is already in the model: `Property.agency`, `Property.agent`, `moderation`, `sourceType`, `ImportJob`, and agency-aware queries throughout. Track B adds only:

- Roles `agency_admin` and `agency_agent` in a single reusable `tenant` access function in `/payload/access/tenant.ts`, applied to every collection.
- Agencies cannot publish directly — saving sets `moderation = unreviewed` and a review queue at `/admin/review` with automated checks (threshold including `internalValueEur`, ≥ 8 images at ≥ 2000 px, description ≥ 600 characters, no contact details in public text, coordinates plausible for the stated market, EXIF stripped) and a publication preview.
- Bulk CSV/XLSX import with a **dry-run validation report** (row-by-row, column-level reasons), idempotent on `(agency, reference)`, images fetched, validated and optimised, `ImportJob` audit, summary email with an error CSV.
- An agency dashboard scoped to their own listings, views, enquiries and response times.
- Playwright tests proving agency A cannot see agency B's listings, enquiries or media through either the UI or the API.

Explicitly not in Track B: automated feeds into the off-market channel. Confidential inventory is entered deliberately, by a person.

10. Design system

10.1 Brand constant

`/src/config/brand.ts` is the only place the brand is named:

```
export const brand = {
  name: 'Lawrence Private Collection',
  short: 'Lawrence',
  legalName: '-', // client to supply
  domain: 'lawrenceprivatecollection.com',
  tagline: 'Exceptional property, openly and otherwise.',
  deskEmail: 'desk@', memberEmail: 'members@', privacyEmail: 'privacy@',
  registeredAddress: '-',
  social: { instagram: '', linkedin: '' },
} as const;
```

Every UI string, email, PDF, JSON-LD and metadata template reads from it.

10.2 Tokens — `/src/tokens/tokens.ts`

```
export const tokens = {
  color: {
    obsidian: '#0A0B0D', // dark sections, headings
    ink: '#15171A', // body text
    graphite: '#4A4D52', // secondary text
    mist: '#9A9A94', // labels, tertiary
    bone: '#F3F0EA', // page surface
    vellum: '#FAF8F4', // raised surface
    line: `rgba(10,11,13,0.10)`,
    patina: '#7E6B4F', // the single warm accent, max 2% of surface
    patinaSoft: '#B9A888',
    focus: '#7E6B4F',
    success: '#3C6B52', warning: '#8A6A24', danger: '#8C3A2E',
  },
  font: {
    display: '"Canela", "Cormorant Garamond", ui-serif, Georgia, serif',
    body: '"Inter", ui-sans-serif, system-ui, sans-serif',
    mono: 'ui-monospace, "SF Mono", monospace',
  },
  size: {xs: '0.75rem', sm: '0.875rem', base: '1rem', lg: '1.25rem', xl: '1.625rem',
    '2xl': '2.25rem', '3xl': '3rem', '4xl': '4.25rem', '5xl': '6rem'},
  tracking: {label: '0.14em', display: '-0.02em'},
  space: [0, 4, 8, 12, 16, 24, 32, 48, 64, 96, 128, 192],
  radius: {none: '0', sm: '2px', md: '3px', lg: '6px'},
  shadow: {card: 'none', pop: '0 12px 48px rgba(10,11,13,.18)'},
  z: {map: 10, sticky: 20, header: 30, drawer: 40, modal: 50, toast: 60},
  bp: {sm: '640px', md: '768px', lg: '1024px', xl: '1280px', '2xl': '1536px'},
  motion: {fast: '200ms', base: '320ms', slow: '420ms', ease: 'cubic-bezier(.22, .61, .36, 1)'},
} as const;
```

Acceptance: changing `color.obsidian` and `font.display` restyles the site, the PDFs and the emails with no other edit.

10.3 Art direction

Full-bleed imagery at ≥ 78 vh; one uniform colour treatment (`saturate(0.98) contrast(1.02)`); fixed aspect ratios (hero 21:9 desktop / 4:5 mobile, cards 4:3, gallery 3:2) so CLS is zero by construction; gradient scrims for legibility; tabular figures for all prices and measurements; letterspaced small-caps labels; margins never below 32 px mobile / 96 px on large screens; no card shadows — separation comes from rules and space; motion 300–420 ms with `prefers-reduced-motion` honoured; fonts self-hosted, subset, max four files.

10.4 The Off-Market invitation state

Where an off-market listing would appear in a public context, render a single quiet line — small-caps label, a count, one link: *"41 properties are held off-market. Create an account to view them."* Never a blurred card, never a lock icon, never a modal that interrupts reading. On an off-market listing page, members see a calm one-line notice that the property is not publicly marketed and asking them to treat the details as confidential.

10.5 Component inventory

Button ▪ QuietLink ▪ Input ▪ Select ▪ Combobox ▪ RangeInput ▪ Checkbox ▪ Switch ▪ Tooltip ▪ Dialog ▪ Drawer ▪ Tabs ▪
 Accordion ▪ Container ▪ Section ▪ Header ▪ Footer ▪ LocaleSwitcher ▪ CurrencySwitcher ▪ UnitSwitcher ▪ PropertyCard ▪
 PropertyCardCompact ▪ Gallery ▪ FactTable ▪ ProvenanceBlock ▪ MarketStatStrip ▪ MapLocality (circle/polygon) ▪
 MapPrecise (staff only) ▪ OffMarketInvite ▪ SaveButton ▪ JoinForm ▪ LoginForm ▪ RequirementsForm ▪ EnquiryForm ▪ DeskCta ▪
 MemberNav ▪ Breadcrumbs ▪ Pagination ▪ Skeleton ▪ EmptyState ▪ Toast ▪ SampleBadge . All rendered in `/dev/styleguide` in
 every state including focus, error, loading and empty.

11. Page specifications

11.1 Home — SSG + ISR 300 s

- Hero** — one full-bleed image, display headline, one line of subcopy, two quiet actions: "View the collection" and "See the off-market properties". No search bar on the homepage; this is not a search product at first contact.
- The Collection** — 6 featured properties, large 4:3 cards showing price (or band), market, headline facts. Never more than three competing for attention on desktop.
- Off-Market** — a short editorial panel with the real live count and a single action to create an account. This is the highest-value block on the page; give it a full viewport.
- Markets** — 8 market tiles, each with one live statistic and its `asOfDate`.
- Intelligence** — the latest report with its ungated summary.

6. **How it works** — three lines: browse openly, create a free account for the off-market collection, speak to the desk.
7. **For owners and brokers** — the supply-side entry.
8. Footer: markets, intelligence, legal, locale/currency/unit switchers, desk contact.

11.2 Collection browse — SSG + ISR

A curated grid with controls that map to **canonical paths, not query strings**: market, property type, price band, segment. Sorting: editorial, newest, price. Optional map view showing locality circles. Deeper filtering (beds, features, area) uses query parameters that are `noindex`, `follow` and never linked from navigation (§15.3). 24 per page, `Pagination` with page 1 canonical.

11.3 Listing detail (public) — SSG + ISR 600 s

Gallery (first image `priority`, lightbox code-split, keyboard and swipe, blurhash) → headline (title, market label, price per `priceDisclosure`, reference, save button) → fact table (beds, baths, receptions, built area, plot in m² and ha, year, architect, heritage status, tenure) → highlights → full description and provenance → **locality map with a circle, never a pin** → features grid → **market context strip** (three live statistics from the `Market` record, each with source and date) → comparable properties in the same market and band → member-only extras block where applicable ("12 further images and the floor plans are available to members") → enquiry form and desk contact → brochure PDF → breadcrumbs → JSON-LD (`RealEstateListing`, `Offer/PriceSpecification`, `BreadcrumbList`) → view counter, server-side, session-deduplicated, non-blocking.

States to design: available, reserved, under offer, sold (90 days, never with a price), expired (410), sample (badge and banner).

11.4 Off-Market section — dynamic, `noindex`, never cached

Index: a dense, scannable grid-and-table hybrid with filters (market, type, budget band, features, "matches my requirements"), served from the **member Typesense collection** using a server-side key that never reaches the browser. A calm one-line notice explains that these properties are not publicly marketed.

Detail: everything a public listing shows, plus documents, the full gallery including member-only assets, running costs and ownership structure where supplied, and three actions — enquire, save, add to requirements. Every view writes `MemberActivity`.

11.5 Market pages — the SEO engine

Editorial introduction answering the query in its first 60 words → **market statistics table**, every figure with source and `asOfDate` → available listings in that market → the off-market count with the join action → buying notes (structures, typical costs, timelines — factual, sourced, never advice) → related markets and segments, 6–10 internal links → FAQ with `FAQPage` JSON-LD → cross-link to the relevant report.

11.6 Intelligence

Index plus report pages. Every report has an **ungated 400–600-word summary with the key figures and charts** and a gated full PDF that requires an account. The summary is the acquisition asset — it is what ranks and what assistants cite; the PDF is what converts a reader into a member. This is the single most efficient registration driver on the site, more than the off-market listings themselves.

11.7 Account

Profile and preferences · saved listings with private notes · requirements profile · email preferences · delete account. Show members their own activity if it is useful, never anyone else's.

11.8 Others

Join · Login · Sell (broker submission and a discreet owner-direct form routed to the desk, never auto-published) · About · Discretion (a genuine page explaining how off-market inventory is handled — it converts) · Contact with Cal.com scheduling · Legal · designed 404 and 410.

12. Copy deck (use verbatim; translate per §13.9)

12.1 Home

- **H1:** "Exceptional property, openly and otherwise."
- **Sub:** "Lawrence Private Collection publishes the world's finest homes from €20 million. The properties that cannot be published are kept for members."
- **Actions:** "View the collection" · "See the off-market properties"
- **Off-market panel:** "Held off-market" / "{count} properties are not publicly listed. Sellers at this level often prefer discretion. Create a free account to view them." / "Create an account"
- **How it works:** "Browse openly — the collection is public, in full." · "Create an account — an email address is all it takes; the off-market properties open immediately." · "Speak to the desk — we arrange viewings and introductions directly with the listing agent."

- **Owners:** "If you are selling and would prefer the market not to know, speak to the desk."

12.2 Listing

- Price: exact, or "Guide €{min}–{max} M", or "Price on request"
- Member extras: "{n} further images and the floor plans are available to members." / "Create a free account"
- Location line: "{Locality}, {Market}, {Country}" with "Exact location provided on enquiry."
- Off-market notice: "This property is not publicly marketed. Please treat the details as confidential."
- Sold: "This property has been sold." (never with a price)
- Expired: "This property is no longer available." + link to the market page
- Sample: "Sample listing — demonstration data, not a real property."
- Enquiry: "Request details" / "The desk replies within one business day." / Consent: "I agree that my details are shared with the listing agent and processed as described in the Privacy Policy."

12.3 Account creation (tone: an open door, not a gate)

- Join H1: "Create an account"
- Sub: "It takes an email address. You will see the off-market collection immediately, and you can save properties and tell us what you are looking for."
- Field labels: "Email" · "Password" · "Or continue with a one-time link"
- Optional block: "Optional — helps us show you the right properties: name, telephone, country, what you are looking for."
- Consent: "Send me new properties matching my requirements." (unticked, optional)
- Confirmation email subject: "Confirm your email to open the off-market collection"
- Welcome: "Your account is open." / "The off-market collection is now visible. Save anything you would like to revisit, and tell us what you are looking for so we can alert you when something arrives."
- Requirements prompt: "Tell us what you are looking for" / "Budget, markets and type. Two minutes, and you will hear first when something matches."
- Match email subject: "A property matching your requirements has arrived"

12.4 Meta templates

- Listing: {PropertyType} for sale in {Locality}, {Market} – {priceOrOnRequest} | Lawrence Private Collection · description: {bedrooms} bedrooms, {builtAreaSqm} m² on {plotArea} in {Locality}, {Market}. {Architect or heritage note}. Trophy property from €20M.
- Market: Prime property for sale in {Market} – market data and homes from €20M | Lawrence Private Collection · description: Prime entry {primeEntry}, {yoy} year on year as of {asOfDate}. {count} properties published, plus off-market inventory for members.
- Segment: {SegmentPlural} in {Market} – {count} for sale | Lawrence Private Collection
- Report: {Title} – Lawrence Intelligence

12.5 Emails (React Email templates)

Confirm your email · welcome · password reset · magic link · new match for your requirements · saved-listing weekly digest (opt-in) · enquiry received (to member) · enquiry notification (to desk and agent) · viewing request · report download link · account deleted confirmation · listing expiring reminder (staff) · weekly desk digest (staff). Token-styled, plain-text fallback, one call to action, logo only.

12.6 Legal pages

Privacy policy · cookie policy · terms of use · member terms (confidentiality expectation for off-market details, no onward disclosure, no scraping) · agency terms (Track B) · listing disclaimer shown on every property page: *"Details are provided by the listing agent. Measurements, boundaries, structures and availability are indicative and must be verified independently before any transaction."*

13. Content sourcing and production — where everything comes from

This chapter exists so the developer never has to ask "what do I put here?". **Nothing in this build waits on the client.**

13.1 Content inventory

Asset	Quantity for launch	Source	Effort
Listing photography	~500 images	Unsplash / Pexels API via the generator (§13.2)	Automatic
Listing data	45 listings (30 public, 15 off-market)	Generator (§13.10)	Automatic
Listing descriptions, 6 locales	270 texts	Template grammar (§13.8)	Automatic
Market pages	18 × 400 words + statistics	AI-drafted from named data sources (§13.6, §13.7)	~3 days
Segment pages	~30 × 300 words	Same	~2 days
Intelligence reports	3 × (600-word summary + 8-page PDF)	Same, built on open data (§13.6)	~2 days
Journal articles	8 × 900 words	AI-drafted, developer-reviewed (§13.7)	~1.5 days
UI strings, 6 locales	~450 keys	Copy deck §12 + machine translation (§13.9)	~0.5 day
Geodata (markets, polygons, coordinates)	18 markets	GeoNames + Natural Earth + OSM (§13.5)	~0.5 day
Fonts, icons	2 families, ~40 icons	Google Fonts + Lucide (§13.3)	1 hour
Logo / wordmark / favicon	1 set	Type-only recipe (§13.4)	2 hours
Agency and agent identities	8 + 16	Fictional, generated (§13.2.4)	Automatic
Legal pages	5	Generator templates, flagged (§13.11)	2 hours
Emails	13 templates	Copy deck §12.5 + React Email	~1 day
FX rates	daily	ECB via Frankfurter API	1 hour

Total content effort: **roughly two weeks inside the build**, with zero client input. The market pages and reports are the expensive part, and they are also the entire acquisition engine — do not shortcut them.

13.2 Photography

Free, commercially usable, API-accessible:

Source	API	Licence position	Use for
Unsplash	unsplash.com/documentation	Free commercial use; the download-tracking call is required by their API terms	Primary source
Pexels	pexels.com/api	Free commercial use, no attribution required	Fallback and variety
Pixabay	pixabay.com/api/docs	Free commercial use	Third fallback
Openverse	api.openverse.org	Aggregated CC — check each licence individually	Last resort
Wikimedia Commons	Commons API	Mixed, often attribution + share-alike	Editorial illustration of named places only

Paid upgrades if the free imagery ever looks generic: Unsplash+ (~€10/month), Adobe Stock, Getty/iStock, Stocksy (best quality), Shutterstock. Ten Adobe Stock images are enough to lift the six homepage slots and the three report covers — a €100 decision the developer can make alone.

13.2.1 Query sets live in `/src/scripts/image-queries.ts`:

```
export const imageQueries = {
  byMarket: {
    saint_tropez: ['saint tropez villa', 'french riviera estate pool', 'provence coastal villa'],
    cap_ferrat: ['cap ferrat villa sea', 'belle epoque villa riviera'],
    monaco: ['monaco penthouse terrace', 'monte carlo harbour view apartment'],
    lake_como: ['lake como villa', 'como lakefront terrace garden'],
    porto_cervo: ['sardinia luxury villa sea', 'costa smeralda villa'],
    tuscany: ['tuscan estate cypress', 'tuscan villa pool countryside'],
    marbella: ['marbella modern villa', 'andalusian luxury villa pool'],
    ibiza: ['ibiza cliff villa sea', 'white modern villa mediterranean'],
    mykonos: ['mykonos villa infinity pool', 'cycladic luxury villa'],
    london_pcl: ['belgravia townhouse', 'london stucco townhouse facade', 'mayfair interior'],
    gstaad: ['gstaad chalet snow', 'luxury alpine chalet interior'],
    courchevel: ['courchevel chalet', 'ski chalet wood interior fireplace'],
    aspen: ['aspen mountain home', 'colorado luxury mountain house'],
    palm_beach: ['palm beach mansion', 'florida waterfront estate'],
    hamptons: ['hamptons beach house', 'long island shingle estate'],
    dubai: ['dubai palm villa', 'dubai penthouse skyline terrace'],
    geneva: ['lake geneva villa', 'swiss lakefront property'],
    algarve: ['algarve clifftop villa', 'portugal coastal luxury villa'],
  },
  byRoom: {
    interior: ['luxury living room fireplace', 'grand entrance hall marble',
      'designer kitchen luxury', 'master bedroom suite view', 'wine cellar luxury',
      'home cinema luxury', 'marble bathroom luxury'],
    exterior: ['manicured estate garden', 'infinity pool sunset', 'estate gates driveway',
      'terrace dining sea view', 'tennis court estate', 'helipad estate'],
    aerial: ['aerial luxury estate', 'drone coastal villa'],
  },
} as const;
```

13.2.2 Selection rules (implemented as filters, not taste). Reject any image under 2000 px on the long edge, or carrying a watermark, a recognisable face in the foreground, a visible brand or licence plate, or a perceptual hash already used in the same market. Prefer landscape 3:2, daylight, no heavy filters. Compose each gallery as: 1 exterior hero · 2 exterior/grounds · 4-6 interior · 1 aerial · 1 detail — so galleries read as authored rather than random.

13.2.3 Storage discipline. Always persist `credit`, `licence`, `sourceUrl`, `sourceId` on `Media`, render attribution in the gallery footer of sample listings, and call Unsplash's download endpoint. This makes the whole demo defensible in one sentence.

13.2.4 People imagery — a real rule. Never use a stock portrait of a real person as a fictional agent. It puts an identifiable individual in front of a business they have never heard of: a personality-rights and GDPR problem independent of the stock licence. Use monogram avatars generated from initials in brand colours (a 20-line SVG component). Real photographs only for real, consenting staff.

13.2.5 AI-generated imagery. Acceptable for abstract editorial headers, report covers and texture, using a model whose terms permit commercial use (Adobe Firefly is safest on training provenance; Flux and Midjourney are usable on their commercial tiers). **Never for listing photography**, even in demo data — an invented photorealistic house presented as a property is exactly the misrepresentation §13.12 exists to prevent. Record `licence: 'ai-generated'` and the model name if used anywhere.

13.3 Fonts and icons

Recommended pairing (free, self-hosted from Google Fonts): display `Cormorant Garamond` or `Playfair Display`; body `Inter`. Alternatives with more character: `Fraunces`, `Instrument Serif`, `Newsreader`, `Libre Caslon Display`; body alternatives `Geist`, `Public Sans`, `Source Sans 3`.

Paid upgrades if a brand budget appears: `Canela`, `GT Sectra`, `Ogg`, `Tiempos` (display); `Söhne`, `Neue Haas Grotesk`, `ABC Diatype` (body). Licensing is per-domain and per-pageview tier — never self-host a paid font without the web licence, and log the decision.

Download via `google-webfonts-helper` or `fontsource`, subset to Latin + Latin-Extended + Cyrillic with `glyphhanger`, `WOFF2` only, four files maximum. **Icons:** `Lucide` (ISC), or `Phosphor`.

13.4 Logo and favicon without a designer

A type-only wordmark is correct for this category and takes two hours.

1. "LAWRENCE" in the display face, all caps, `letter-spacing: 0.18em`, weight 400, optical size ≥ 48 px; "PRIVATE COLLECTION" beneath in the body face, small caps, `letter-spacing: 0.22em`, at 38 % of the wordmark size.
2. Optional 1 px rule between the two lines at 100 % of the wordmark width.
3. Monogram for favicon and app icons: "L" in the display face on `color.obsidian` with 12 % padding.
4. Export SVG + PNG at 1x and 2x, generate the full icon set with `realfavicongenerator.net`, commit `site.webmanifest`.
5. Everything in `/public/brand/`, referenced only through `/src/config/brand.ts`.

13.5 Geodata — markets, coordinates, polygons

Need	Source	Licence
Cities, regions, coordinates, population, localised alternate names	GeoNames (geonames.org)	CC BY 4.0 — attribute in the colophon
Country and region polygons	Natural Earth	Public domain
Finer market polygons (e.g. Belgravia, Cap Ferrat)	OpenStreetMap via Overpass (overpass-turbo.eu)	ODbL — attribute "© OpenStreetMap contributors" on every map
Geocoding during import	Nominatim (1 req/s policy) or a paid geocoder in bulk	ODbL
Localised country and region names	i18n-iso-countries + CLDR	Open
Map tiles	MapTiler free tier (100k loads) or self-hosted Protomaps .pmtiles	Commercial / BSD

Cache extracts in `/data/geo/` and refresh quarterly. Never query Overpass or Nominatim at runtime.

13.6 Market data — the sources that make the research real

Every published figure needs a **source URL** and an `asOfDate`. These are the sources to use, all free:

Region	Source	What it gives
Italy	Agenzia delle Entrate — OMI (agenziaentrate.gov.it , Osservatorio del Mercato Immobiliare)	Official €/m² quotations by municipality and zone, twice yearly — the single best free property dataset in Europe
Italy	ISTAT	House price index, transaction volumes
UK	HM Land Registry Price Paid Data + UK House Price Index (Open Government Licence)	Every recorded transaction, address-level, free and reusable
France	DVF — Demandes de valeurs foncières (data.gouv.fr) and Notaires de France indices	Transaction prices by commune
Spain	INE + Ministerio de Transportes (MITMA) housing statistics	Price per m², transactions
Portugal	INE Portugal	House price index by municipality
Greece	Bank of Greece residential property price indices	
Switzerland	BFS/OFS and cantonal registries	
USA	FRED (St. Louis Fed), Redfin Data Center , Zillow Research , county assessor portals	County-level medians, days on market
UAE	Dubai Land Department open data	Transaction volumes and prices
Global	BIS residential property price database , Eurostat , OECD	Cross-country comparability
Context	Knight Frank Wealth Report, Savills Research, Christie's Luxury Defined	Cite with attribution and a link — never reproduce tables or paragraphs

The rule that makes this legally and editorially safe: we publish *our own* derived figures from open data, plus short attributed references to published research. We never copy a table, a chart or a paragraph from a commercial report. A sentence such as "Knight Frank reported prime prices in {market} rose x % in {year} (source, date)" is fine; reproducing their index table is not.

Store the figures in the `Market.stats` group with `source` and `asOfDate`, and surface them through `/api/public/markets/{market}/stats` (\$15.6). Refresh quarterly; a stale index is worse than none.

13.7 Editorial copy — the writing brief and the AI workflow

The developer writes the editorial with AI assistance. This is a legitimate, high-quality workflow; unreviewed AI output is not.

Market page (400–450 words): a 40–60-word answer paragraph opening the page (what the market is, what the entry price is, what you get) · one paragraph on character and who buys there · one paragraph of hard specifics with our own figures · one paragraph on buying practicalities (structures, typical costs, timelines, seasonality) · closing line to the listings. Two question-form H2s. No adjective that could apply to any market.

Journal article (800–1,000 words): answer-first opening, three to five question-form H2s, one table or list of concrete figures, every non-obvious claim sourced with a link and a date, a short "what this means for buyers" close.

Report (600-word summary + 8-page PDF): a finding, a method note, three to five charts from open data, a table by market, and a short caveats section. Honest methodology beats impressive claims: state the sample, the source and the limits.

Four non-negotiable rules:

1. Never publish a statistic produced from memory. Every number comes from a named source with a URL and a date, or from our own database aggregates.

2. Every external claim carries an inline source and `asOfDate`.
3. The developer reads every published sentence. Anything not verifiable in 30 seconds is deleted rather than softened.
4. No AI-generated text about law, tax or residency beyond stating that a rule exists and linking the official source.

Article backlog for launch (pick 8): how off-market sales actually work at €20 M+ · buying through an SPV versus directly, and when share transfer makes sense · residency and investment-visa status across Portugal, Greece, Italy, Spain and the UAE (facts + official links only) · what it costs to complete: taxes and fees by market · heritage and listed-building constraints in Italy and France · the real timeline of a €50 M transaction · why prime markets decouple from national indices · running costs of a large estate, itemised · buying a private island: the eight questions to ask first · ski property: altitude, access and the value of ski-in/ski-out · what a helipad or a 40 m berth adds to value · the discretion playbook: selling without the market knowing.

13.8 Listing descriptions — template grammar

Listing copy is generated from a slot grammar per locale in `/src/content/description-templates/{locale}.ts`:

```
{OpeningByType} {PositioningByMarket} {AccommodationSentence} {GroundsSentence}
{ProvenanceSentence?} {FeatureSentence} {LocationSentence} {ClosingByTier}
```

Six to ten variants per slot, selected deterministically by listing id so copy is stable across runs and no two listings read alike. Numbers always come from structured fields, never from prose. This produces plausible, locale-native copy at zero marginal cost — and it is how real agency descriptions get written anyway.

13.9 Translations

English is the source of truth. Draft `it`, `fr`, `de`, `es`, `ru` with the DeepL API or an LLM pass, mark each string `machineTranslated: true`, and hand-check the 40 highest-traffic strings (navigation, CTAs, form labels, error messages, consent text). **Never machine-translate the legal pages or the consent copy.** Keep `/docs/translation-review.md` for a native reviewer later; a professional review of ~450 strings costs roughly €300–500 per language and is a post-launch proposal, not a blocker.

13.10 Sample inventory generator

`/src/scripts/seed-sample-data.ts` creates **45 properties — 30 public and 15 off-market** — across Saint-Tropez, Cap Ferrat & Cap d'Antibes, Monaco, Lake Como, Porto Cervo, Tuscany, Marbella & La Zagaleta, Ibiza, Mykonos, London Prime Central, Gstaad, Courchevel, Aspen, Palm Beach, the Hamptons, Dubai, Lake Geneva and the Algarve. Values €10 M–€180 M, weighted so `trophy` dominates, `signature` is ~20 %, and exactly four `prime` exceptions demonstrate the 10 % valve. Every `priceDisclosure` mode is represented so all three price states can be demonstrated.

Also generated: 8 fictional agencies and 16 agents with monogram avatars; **8 test members** (active, pending, suspended, with and without requirements) whose credentials live only in `.env.local`; 12 saved listings and a populated `MemberActivity` so the dashboards are not empty; 18 `Market` records with clearly-marked sample statistics; 3 reports; 8 journal articles; 30 segment pages.

Deterministic: `@faker-js/faker` with a fixed seed, idempotent upsert on `(agency, reference)`, flags `--count`, `--market`, `--wipe`, and a summary table on completion.

13.11 Legal and policy text

Draft placeholders from a reputable generator (Termly, iubenda, or the GDPR.eu templates) plus a cookie table generated from the cookies actually in use, each file marked `PLACEHOLDER – pending legal review` in a header comment and in an admin note. The developer produces: privacy policy, cookie policy, terms of use, member terms, listing disclaimer. Legal review is a client task at the end and blocks nothing.

13.12 Sample-data safety rules (mandatory)

Publishing invented property adverts as real is a consumer-protection problem in most of our markets and poisons the index before launch. Therefore: every sample record has `isSample = true`; a visible "SAMPLE LISTING — demonstration data" badge and banner in all locales; `noindex`, `nofollow` and excluded from sitemaps while `SAMPLE_DATA_ENABLED=true`; enquiries on sample listings log but never email an agent; sample member accounts never receive real email (route to a catch-all); no real agency, agent, owner, address or telephone number; monogram avatars only, never stock portraits; coordinates plausible for the market but never on an identifiable private property; `pnpm sample:purge` removes every sample record, member, media asset and activity row; the launch checklist blocks go-live while any sample content is publicly reachable.

13.13 Pre-launch content checklist

Green before DNS cutover: hero imagery selected and within budget · all 45 sample listings have ≥8 images and a complete description in every locale · 18 market pages have reviewed editorial copy and at least three sourced statistics each · ~30 segment pages published · 3 reports with ungated summaries and downloadable PDFs · 8 articles with sources · UI strings translated and the top 40 hand-checked · geodata loaded with attribution rendered on maps · fonts subset and self-hosted · logo and favicon set committed · legal placeholders in place and flagged · FX snapshot job running · `audit:exposure` and `audit:seo` clean · sample data purged if real inventory has arrived.

14. Performance

Metric	Public layer	Member area
LCP	< 1.2 s	< 1.8 s
CLS	< 0.02	< 0.05
INP	< 150 ms	< 200 ms
TTFB	< 150 ms (edge)	< 400 ms (dynamic)
First-load JS — home	< 110 KB gz	—
First-load JS — listing	< 130 KB gz	< 170 KB gz
First-load JS — collection (pre-map)	< 150 KB gz	—
Hero image transferred	< 120 KB AVIF	—
Lighthouse mobile perf / a11y / SEO	≥ 95 / 100 / 100	≥ 90 / 100 / n/a
Signed media first byte	—	< 300 ms

Rules. Public pages are SSG or ISR only; `force-dynamic` is forbidden there. The member area is `force-dynamic` with `Cache-Control: private, no-store` and must never be edge-cached. Map, lightbox, charts and the PDF renderer are `next/dynamic` with `ssr:false`. AVIF first with explicit `sizes`, variants at upload never on demand, exactly one `priority` image per page, blurhash elsewhere. Fonts self-hosted, subset, preloaded, max four files. Fixed aspect ratios and dimension-matched skeletons everywhere. ISR plus a signed `/api/revalidate` webhook from Payload `afterChange`. Analytics is the only third-party script and it loads `afterInteractive`; no tag managers, no chat widgets, no A/B tools in Phase 1. CI enforces every budget and fails the build on regression. Perceived speed: prefetch on card hover, optimistic saves, instant skeletons, View Transitions where supported.

15. SEO and AI discovery

15.1 The principle

The public Collection is genuinely open and complete, so it can rank like any strong portal. The research library is what earns authority and citations. The account gate sits only in front of the off-market inventory and the report PDFs — never in front of content we want indexed.

15.2 Foundations

Canonical on every page · hreflang for all 6 locales + `x-default` · sitemap index split by type (properties, markets, segments, reports, journal, static), 50k URLs max per child, correct `lastmod`, excluding off-market, sample, expired and non-public content · `robots.txt` · designed 404 and 410 · one H1 per page · breadcrumbs everywhere · 6–10 contextual internal links per market page · image filenames and alt text generated from structured data (`villa-saint-tropez-sea-view-01.avif`, alt "Nine-bedroom villa with sea view in Saint-Tropez, Côte d'Azur").

15.3 A finite crawl surface

All indexable browse states are **canonical paths** under `/collection/...` and `/markets/...`. Query-string filters are `noindex, follow` and never linked from navigation. Pagination beyond page 1 is `noindex, follow` with a canonical to page 1. `/off-market/*` and `/account/*` are excluded by `robots.txt`, by `X-Robots-Tag`, **and** by authentication — three independent layers.

15.4 The market-intelligence engine (the real growth strategy)

Modelled on Knight Frank and Savills research. Phase 1 ships:

- **18 market pages** with sourced, dated statistics: Saint-Tropez, Cap Ferrat & Cap d'Antibes, Monaco, Lake Como, Porto Cervo, Tuscany, Mallorca, Ibiza, Marbella & La Zagaleta, Mykonos, London Prime Central, Geneva & Lake Léman, Gstaad, Courchevel & Méribel, Aspen, Palm Beach, the Hamptons, Dubai.
- **~30 market × segment pages**, each gated on real editorial copy plus three data points.
- **3 reports** with ungated summaries: a prime-entry-price comparison across markets, a note on purchase structures and total acquisition costs in four jurisdictions, and a study of published versus off-market share by market.
- **A journal** answering what this audience actually searches (\$13.7 backlog).

15.5 Structured data

RealEstateListing + Offer/PriceSpecification (bands as minPrice/maxPrice) · BreadcrumbList everywhere · Organization + WebSite with SearchAction · FAQPage on market pages · Article on journal posts · Dataset on reports and on the market stats endpoint — under-used and ideally suited to a price index · RealEstateAgent on agency pages. Vitest shape tests plus a Rich Results check in CI.

15.6 AI and answer-engine discovery

1. `/llms.txt` — a Markdown map of the site: what it is, the admission threshold, the market index, the data licence, contact. `/llms-full.txt` lists every market and report.
2. **Crawler policy** in `robots.txt`: allow GPTBot, OAI-SearchBot, PerplexityBot, ClaudeBot, Google-Extended on the public layer; disallow `/off-market`, `/account`, `/api`, `/admin` and filter URLs. Default is allow — being the cited source for prime-market data is worth far more than the content.
3. **Answer-first structure** — every market page and report summary opens with a 40-60-word direct answer containing the key figures with units and dates, then question-form H2s.
4. **Machine-readable data** — `/api/public/markets/[market]/stats` returning the published figures as JSON with `source` and `asOfDate`, referenced by Dataset JSON-LD. Cheap to build, disproportionately citable.
5. **Entity clarity** — consistent `Organization` with `sameAs`, a plainly worded About page, consistent market naming across the whole site.
6. **Freshness** — honest `dateModified` in JSON-LD and sitemaps, updated when the data actually changes.
7. **Measurement** — track assistant-referral domains in Plausible and record a monthly citation spot-check in `DECISIONS.md`.

15.7 What must never be indexable

Off-market listings and their ids, member identities, exact coordinates, `addressLine`, `internalValueEur`, commission terms, documents and signed media URLs. `pnpm audit:exposure` asserts all of it on every build — the most important script in the repository.

16. Accessibility — WCAG 2.2 AA

This audience skews older and uses large screens and, increasingly, assistive technology. Accessibility here is commercial, not only legal — and in the EU the European Accessibility Act now reaches services of this kind.

Contrast $\geq 4.5:1$ for body text and $\geq 3:1$ for large text and UI borders — check `mist` and `patina` against `bone` before shipping · every interactive element a real `button` or `a`, keyboard-reachable, with a visible 2 px `focus-visible` ring · gallery with arrow keys, `Esc`, focus trap and restoration, announced image counter · a "skip map" link and a non-map equivalent for every map action · labelled inputs (labels, not placeholders), `aria-live` result counts, error summaries focused on submit, errors linked with `aria-describedby` · target size $\geq 24 \times 24$ px · `prefers-reduced-motion` honoured · logical heading order · **the account creation and login flow must be completable with a keyboard and a screen reader** — a gate that excludes is a gate that loses clients. axe-core on 10 routes in CI including the member area, plus a recorded manual keyboard pass.

17. Security and privacy

17.1 Application security

Member sessions in `HttpOnly`, `Secure`, `SameSite=Lax` cookies with a 30-day remember-me and silent refresh; optional TOTP 2FA for members, **mandatory for staff**; rate limits on login (5/15 min), join (3/h/IP), enquiry (5/h), signed media (60/min/member), password reset (3/h); Turnstile on join, login and enquiry; strict CSP compatible with MapLibre and Plausible; HSTS, `Referrer-Policy: strict-origin-when-cross-origin`, `Permissions-Policy`, `X-Content-Type-Options`; signed webhooks for revalidation and cron; dependency audit in CI; Sentry with PII scrubbing and **no request bodies captured on member routes**.

17.2 Exposure integrity (the critical control)

Allowlist projections (\$8.3) · `audit:exposure` in CI · Playwright tests fetching every public route anonymously and asserting the absence of off-market markers · a test asserting that an anonymous request to an off-market id returns 404 (not 403 — never confirm existence) · a test asserting signed media URLs expire and are single-use · a test asserting `internalValueEur` and `addressLine` never appear in any response body. Treat a failure here as a production incident, not a test failure.

17.3 Multi-tenancy (dormant but enforced)

Even in single-team Phase 1, every listing query is agency-aware and the `tenant` access function exists with tests. This is what makes Track B a configuration change rather than a refactor.

17.4 Privacy

Phase 1 processes ordinary contact data only — no identity documents, no financial documents — which keeps the compliance burden light and a DPIA unnecessary at this stage (it becomes necessary if §23.2 is ever built). Requirements: non-blocking, deferred, granular cookie consent with a stored consent record; analytics only after consent where required; privacy, cookie and terms pages live before launch; DPAs with Vercel, Neon, Cloudflare, Upstash, Resend, Typesense, Sentry, Cal.com; a processing register; self-service account deletion plus a one-click admin anonymisation that preserves pseudonymised `MemberActivity`; a documented subject-access procedure in `/docs/privacy-procedures.md`.

17.5 Retention

Enquiries 24 months then anonymised · `MemberActivity` 24 months · dormant member accounts flagged at 24 months and deleted at 36 after two notices · staff `AuditLog` 24 months · sample data purged before launch.

17.6 Anti-scraping

No public bulk endpoint; unguessable media paths; rate limits and Cloudflare bot rules; no exact coordinates in any public response; the off-market channel requires a session plus a signed URL. State the prohibition in the Terms. Accept that the public layer can be copied — it is deliberately the part we are content to have copied.

18. Analytics

Public layer — Plausible, events defined once in `/src/lib/analytics.ts`: `collection_viewed` · `listing_viewed` (id, market, band) · `filter_applied` · `off_market_cta_clicked` (from which surface — this tells you exactly what drives registration) · `join_started` / `join_completed` · `report_summary_read` · `report_download_gated` · `market_viewed` · `enquiry_submitted` (source) · `desk_call_scheduled` · `save_clicked` · `locale_switched` · `currency_switched`. No personal data in payloads.

Member area — no third-party analytics at all. Product metrics come from `MemberActivity` aggregates in the admin: active members, off-market views per listing, document downloads, saves, requirement matches proposed and acted on, and the funnel from first visit to account to enquiry. Nothing personal leaves the system.

19. Testing

Unit (Vitest): audience projections — a table-driven test enumerating every viewer × channel × field group and asserting exact presence or absence (the highest-value test in the codebase) · filter-to-Typesense translation · price-band formatting and FX conversion · signed-URL generation and expiry · combo/segment resolver and canonicalisation · metadata and JSON-LD builders · match scoring · description template determinism.

E2E (Playwright): join → confirm email → off-market opens, end to end · an anonymous request to an off-market id returns 404 · off-market listings absent from sitemap, OG routes, public search and related-listing suggestions · signed media URL fails after expiry and on second use · public listing renders correctly for all three `priceDisclosure` modes · save, note and requirements flows persist · enquiry submits, routes and emails · expired listing returns 410 · sample content emits noindex and is absent from sitemaps · locale switch preserves path and query · `audit:exposure` passes · keyboard-only completion of join and enquiry · (Track B) agency A cannot see agency B's listings or enquiries.

Non-functional: Lighthouse CI budgets, axe-core on 10 routes, bundle-size assertions, visual regression on 10 viewports.

20. Runbooks, Definition of Done, launch

20.1 Runbooks

Deploy: merge to `main` → CI gates → Vercel promotion; instant rollback, plus a committed down-migration for every schema change. **Database:** Neon daily backups with 7-day PITR; migrations reviewed as SQL before applying. **Search:** reindex into an alias and swap, never in place; the member collection is written with the admin key only, never from a client; if Typesense is unreachable, a Postgres fallback query path serves results with reduced faceting — implement it, do not assume it. **Exposure incident:** if off-market data is suspected to have leaked, switch the affected listings to `draft`, pull `MemberActivity` for them, check CDN and search-engine caches, request removal via Search Console, notify the seller within 24 h, document in `DECISIONS.md`, then fix the code. **Content freeze / launch:** 48 h before cutover freeze content edits, verify sitemaps and redirects, purge sample data, set `SAMPLE_DATA_ENABLED=false`, then cut DNS with a 300 s TTL set 24 h earlier. Availability target 99.9 %.

20.2 Definition of Done

1. Every budget in §14 verified in Lighthouse CI and on PageSpeed Insights.

- 2. Public Collection complete: browse, listing detail in all three price modes, all lifecycle states, comparable listings, brochure PDF.
- 3. Off-Market works end to end: join, email confirmation, immediate access, saved listings, requirements, match email — and `audit:exposure` is green.
- 4. Member media and documents served only via signed URLs with expiry and single use proven by test.
- 5. CMS complete: an editor can create, localise and publish a listing, and set its publication mode, without a developer.
- 6. 45 sample listings, 8 sample members and populated activity live, badged, noindex, purgeable.
- 7. 18 market pages and ~30 segment pages published with sourced, dated statistics; 3 reports with ungated summaries; 8 articles; `llms.txt` served.
- 8. Valid JSON-LD on every template; `audit:seo` clean; sitemaps exclude everything gated or sample.
- 9. WCAG 2.2 AA on 10 routes including join and the member area.
- 0. Security headers grade A, rate limits verified, CSP clean, staff 2FA enforced.
- 1. Enquiries save with consent, route correctly, notify, and are rate-limited.
- 2. Changing `/tokens/tokens.ts` restyles site, PDFs and emails.
- 3. `/lib/services` empty and documented; every §23 seam present and unused.
- 4. `DECISIONS.md`, `README.md` (new developer running in 10 minutes), `/docs/admin-guide.md`, `/docs/architecture.md`, `/docs/privacy-procedures.md` complete.
- 5. Staging signed off; launch checklist executed; `SAMPLE_DATA_ENABLED=false` in production; §13.13 content checklist green.

21. Pre-answered decisions

#	Question	Default
1	Brand name	Lawrence Private Collection; short form "Lawrence"
2	Admission threshold	€20 M, with a <code>prime</code> exception track €10–20 M capped at 10 % of published inventory
3	Below-threshold submissions	Rejected automatically with a courteous explanation
4	On-request pricing	Allowed; <code>internalValueEur</code> still required for threshold enforcement
5	Default price display	As the seller instructs: exact, band, or on request
6	Off-market gate	Free account, email confirmation only. No NDA, no proof of funds, no vetting
7	Manual approval of members	Off by default; <code>MEMBER_REQUIRE_APPROVAL</code> flag exists
8	Social login	No — it discloses membership to a third party
9	Member 2FA	Optional for members, mandatory for staff
10	Locales	en (default), it, fr, de, es, ru
11	Currencies	EUR base; USD, GBP, CHF, AED, SGD, HKD display, daily ECB snapshot
12	Units	Metric default, imperial toggle; land shown in both hectares and acres
13	Public map precision	Locality circle, never a pin; exact coordinates staff-only
14	Minimum media	8 images at ≥2000 px; EXIF always stripped
15	Watermarking	None in Phase 1 (seam in §23.3)
16	Documents	Members only, signed URLs, 15-minute expiry
17	Listing freshness	120 days, T-14 reminder, then expired + 410
18	Sold listings	90 days without a price, then noindex, then 301 to the market page
19	Duplicate listings	Canonical listing displayed, duplicates linked
20	Featured slots	6 on home, editorial choice only
21	Default sort	Editorial, then newest
22	Pagination	24 per page, page 1 canonical, later pages noindex follow
23	Report gating	Summary open, PDF requires an account
24	Newsletter	Opt-in at join and at report download, double opt-in, digest sending in Phase 1 limited to match emails
25	Match emails	Opt-in, maximum one per day per member
26	Enquiry routing	Desk first, then listing agent; response commitment one business day
27	Live chat	Never

28	Desk contact	Enquiry form plus Cal.com scheduling; no published direct number in Phase 1
29	Agency accounts	Track B, after launch unless the client has agency inventory waiting
30	Automated feeds	Public channel only, never off-market, and only in Track B
31	Auto-approval of listings	Not applicable in Phase 1 (single team); in Track B, never
32	AI crawlers	Allowed on the public layer, blocked on /off-market , /account , /api , /admin
33	Cookie banner	Non-blocking, granular, one-click reject
34	Analytics	Plausible on the public layer only
35	Domain strategy	Single domain, locale paths, apex canonical, www 301s, no trailing slash
36	Error pages	Designed 404 and 410 with markets and the collection
37	Session length	30 days remember-me, silent refresh, logout everywhere available
38	Signed URL lifetime	15 minutes, single use
39	Account deletion	Self-service, immediate, anonymises activity
40	Sample data	Purged before DNS cutover
41	Uptime target	99.9 %, monitored externally
42	Support addresses	desk@, members@, agents@, privacy@
43	Blog authorship	House byline "Lawrence Intelligence" until real authors exist
44	Market data refresh	Quarterly, with asOfDate shown on every figure
45	Image source	Unsplash/Pexels for sample data; agency-supplied for real listings
46	Agent avatars	Monograms for sample data; real photos only for real, consenting people
47	Legal text	Generator placeholders, clearly flagged, lawyer review before go-live
48	Handover	README, DECISIONS, admin guide, architecture doc, privacy procedures, launch checklist, recorded walkthrough

22. READY-TO-PASTE CURSOR PROMPTS

In order. Track A is Prompts 1-14; Track B is 15-16 and can run before or after launch. Each prompt ends with acceptance criteria — require a pass/fail report per line before committing. Keep `CLAUDE.md`, §6 and §8 open as context.

Prompt 1 — Scaffold and guardrails

```
Set up the project "lawrence":
- Next.js 15 App Router + TypeScript strict (noUncheckedIndexedAccess), Payload 3 embedded with the
  Postgres adapter, Tailwind v4 reading CSS variables, ESLint + Prettier.
- Folder structure exactly as spec §7.4, with route groups (public) and (member). Every route under
  (member) is force-dynamic with Cache-Control: private, no-store and X-Robots-Tag: noindex, applied
  once in a shared layout – not per page.
- Zod-validated env module failing fast at boot; .env.example with every variable in §7.3.
- /src/config/brand.ts exactly as in §10.1 – the only place the brand is named.
- /api/health returning { ok, db, search, redis, version }.
- docker-compose with postgres and typesense; a migration enabling postgres and pg_trgm.
- Vitest + Playwright + axe + Lighthouse CI; GitHub Actions running typecheck, lint, test, e2e, lhci,
  bundle budgets, audit:seo, audit:ally and audit:exposure on every PR.
- CLAUDE.md from spec §1.1, /.cursor/rules/lawrence.mdc, empty DECISIONS.md.
- Scripts: dev build start typecheck lint test test:e2e analyze seed sample:purge search:reindex
  audit:seo audit:ally audit:exposure audit:all.
ACCEPTANCE: pnpm dev boots; /api/health ok; CI green; a new route added under (member) is
automatically noindex and uncached, proven by a curl assertion in a test.
```

Prompt 2 — Tokens and components

Build the design system per spec §10, using the exact token values in §10.2.
Generate CSS variables from tokens, wire Tailwind v4 via @theme, and add an ESLint rule banning hex colours, px font sizes and arbitrary spacing outside /src/tokens.
Self-host Cormorant Garamond (display) and Inter (body) per §13.3: subset Latin + Latin-Extended + Cyrillic, WOFF2 only, max 4 files, font-display swap, preload only the above-the-fold display weight.
Use tabular figures for every price and measurement.
Build the component inventory in §10.5 using tokens only, including OffMarketInvite rendering exactly the state described in §10.4 (small-caps label, real count, one quiet link – never a blur, never a lock icon) and SaveButton with an optimistic state.
Fixed aspect-ratio wrappers on every image component.
/dev/styleguide (dev-only) renders every component in every state.
ACCEPTANCE: changing color.obsidian and font.display restyles the whole styleguide; axe reports zero violations; every control is keyboard reachable with a visible focus ring; contrast verified for mist and patina on bone.

Prompt 3 — Property collection

Create the Payload collection Property exactly per spec §6.1–6.5, localised into en (default), it, fr, de, es, ru. Include ALL fields, including agency and agent (present from day one even though Phase 1 is single-team), channel, valueTier, moderation, internalValueEur, priceDisclosure, priceBandMin/Max, ownershipStructure, saleStructure, provenance, heritageStatus, the full feature enum, the optional waterfront sub-block, and the location group with coordinatePrecision and publicGeography.

Hooks:

- beforeValidate: block publication when internalValueEur (or priceAmount) is below €20M unless valueTier = prime AND an admin override is set; enforce the 10% cap on prime listings in the public channel and explain any rejection in plain language.
- beforeChange: compute priceEur from the stored daily FX snapshot; derive valueTier; generate slug only for channel = public; set expiresAt = publishedAt + 120 days; compute fingerprint.
- afterChange: upsert into the correct Typesense collection (public_listings vs member_listings) based on channel; trigger ISR revalidation for public listings only.

Admin UI: field groups ordered Essentials → Publication → Commercial → Physical & provenance → Location → Media → Documents → SEO → Publishing, with the single "How should this property be published?" control from §8.4 and a live preview of what an anonymous visitor and a member see. Drafts + versions + autosave. All indexes in §6.8. Show me the SQL migration before applying.

ACCEPTANCE: a €12M listing cannot publish without the prime override; an off-market listing never receives a slug; Typesense routing is correct for both channels; types compile; EXPLAIN shows index usage on the hot public query.

Prompt 4 — Remaining collections

Create: Media (variants at upload per §14, EXIF stripped without exception, 2000px minimum enforced with a clear error, blurhash, per-asset visibility public|members routing to the public or private bucket, credit/licence/sourceUrl, agency scoping), Document (members only, private bucket), Member, SavedListing, Requirement, MemberActivity (append-only – no update or delete for anyone, including admins), Enquiry, Market (with the stats group where every figure carries source and asOfDate), Report, Article, Page, Taxonomy, Agency, Agent, Redirect, User (admin/editor now; agency roles reserved), AuditLog, ImportJob, FxSnapshot – all per spec §6.5–6.7.

Member auth: Payload auth on the Member collection, email + password and magic link, email confirmation required before status becomes active, optional TOTP 2FA, MEMBER_REQUIRE_APPROVAL flag routing new accounts to pending when true.

Write the RESERVED unused fields on Member (ndaStatus, ndaSignedAt, capabilityStatus, capabilityBand, tier) so §23 is a flag later, not a migration. Never write to them now.

ACCEPTANCE: a member cannot read another member's record through any API; MemberActivity rejects updates and deletes; an unconfirmed account cannot reach any (member) route; document uploads land only in the private bucket and are unreadable without a signed URL.

Prompt 5 — The access layer (before any page that renders listing data)

Implement spec §8.1–8.3.

- /src/lib/access/viewer.ts resolving the Viewer from the request (anonymous, member with status, staff, agency) – no ambient globals.
- /src/lib/access/visibility.ts implementing canSee() exactly as specified.
- /src/lib/access/projections.ts: ALLOWLIST projections per audience, exactly the table in §8.3, including the never-exposed fields (internalValueEur, commissionTerms, addressLine, exact coordinates, owner data).
- Every function in /src/lib/db takes Viewer as its first argument and applies the projection before returning. Make it impossible to compile a call without a viewer.
- A table-driven Vitest suite enumerating EVERY viewer × channel × field group with the exact expected presence or absence, no skipped cells.
- /src/scripts/audit-exposure.ts implementing the audit described in CLAUDE.md, wired into CI as a blocking check.

ACCEPTANCE: the enumerated table test passes; audit:exposure passes; removing a field from a projection makes a test fail rather than silently leaking; adding a new model field exposes it nowhere by default.

Prompt 6 — Data layer, geo, search

```
Build /src/lib/db (getListingBySlug, getOffMarketById, getFeatured, getComparables, searchListings,
getMarketAggregates, getMarketStats, getAdminDashboardStats, getMemberDashboard), /src/lib/geo
(bbox, radius, ST_DWithin helpers, and a deterministic jitter used for the locality circle so a
public listing never exposes an exact point) and /src/lib/search (two Typesense collections –
public listings and member listings – with distinct keys; only the public search-only key may reach
the browser; full reindex into an alias with atomic swap; incremental upsert/delete from the Property
hooks; a Postgres fallback path used automatically when Typesense is unreachable).
ACCEPTANCE: unit tests cover filter translation; no public API response contains exact coordinates;
grepping the client build finds no member search key; killing Typesense locally still returns results.
```

Prompt 7 — i18n, currency, units

```
Set up next-intl for en (default), it, fr, de, es, ru per spec §5.1.
Per-locale routing, language detection only on "/" (302, uncached), all other routes always prefixed,
no trailing slash. /src/messages/*.json seeded with the complete copy deck from §12 in English, with
machine-translated drafts for the other locales marked machineTranslated:true per §13.9. Fallback to
en with a dev-time warning listing missing keys. LocaleSwitcher preserving path and query and mapping
localised market and segment slugs. CurrencySwitcher converting from priceEur using the stored daily
FX snapshot, showing the original currency on hover. UnitSwitcher (metric/imperial, land in ha and
acres) stored in a cookie. One hreflang helper emitting all locales + x-default on every page.
ACCEPTANCE: switching locale on a market page lands on the equivalent page; hreflang validates; an
ESLint rule proves no literal user-facing string exists in any component.
```

Prompt 8 — Public Collection pages

```
Build the public layer per spec §11.1–11.3 and the copy deck §12.1–12.2: home, collection browse
(canonical paths only – no crawlable filter query strings), and the listing detail with all three
priceDisclosure modes, the locality circle map, the market context strip, comparable listings, the
member-extras block, the enquiry form, the brochure link, breadcrumbs and JSON-LD per §15.5.
SSG + ISR, dynamic OG images, and the available / reserved / under offer / sold / expired / sample
states. Expired returns HTTP 410 with the designed page.
ACCEPTANCE: Lighthouse mobile ≥95 with LCP < 1.2s on home and listing; first-load JS within the §14
budgets; audit:exposure passes against an anonymous crawl of every generated page; no response for
any public listing contains exact coordinates, addressLine or internalValueEur.
```

Prompt 9 — Membership and the Off-Market section

```
Implement spec §8.5–8.8 and §11.4 with the copy in §12.3.
- /join: email + password or magic link, optional fields, optional marketing consent, Turnstile.
  Email confirmation opens the account immediately. MEMBER_REQUIRE_APPROVAL defaults to false.
- Login, logout, password reset, optional TOTP enrolment.
- /off-market and /off-market/[id]: dynamic, noindex, never cached, served from the member Typesense
  collection with a server-side key only. Filters per §11.4 including "matches my requirements".
- Member media and documents through /api/secure/media/[token] and /api/secure/document/[token]:
  HMAC over (assetId, memberId, expiry, nonce), 15-minute lifetime, nonce burned in Redis, session
  re-checked server-side, private no-store headers, and 404 (never 403) for unauthenticated requests.
- MemberActivity written on off-market list views, detail views, document downloads, saves and
  enquiries.
- Account area: profile, saved listings with private notes, requirements profile, email preferences,
  self-service deletion that anonymises activity.
- /src/lib/match.ts scoring requirements against new listings, plus the match email (opt-in, max one
  per day per member) and an admin matching board.
ACCEPTANCE: the join → confirm → off-market journey passes end to end in Playwright; an anonymous
request to an off-market id returns 404; a signed URL fails after expiry and on second use; off-market
ids appear in no sitemap, OG route, public search response or related-listing block; deleting an
account leaves no personal data and preserves pseudonymised activity.
```

Prompt 10 — Market intelligence and the SEO engine

```
Build /markets, /markets/[market], /markets/[market]/[segment], /intelligence and
/intelligence/[slug] per spec §11.5, §11.6 and §15.4, plus /journal.
Every statistic renders with its source link and asOfDate. Reports have an ungated 400–600 word
summary with charts and a gated PDF requiring an account.
Seed 18 market records and ~30 segment pages with editorial copy written per the brief in §13.7,
answering the query in the first 60 words, using figures drawn from the open sources listed in §13.6
with a source URL and date on every number. Seed 3 reports and 8 articles.
Implement the full SEO layer: the metadata helper with the templates in §12.4, canonical, hreflang +
x-default, dynamic OG images, a sitemap index excluding off-market, sample, expired and gated
content, robots.txt with the AI crawler policy in §15.6, /llms.txt and /llms-full.txt, JSON-LD per
§15.5 including Dataset on reports, and /api/public/markets/[market]/stats returning sourced JSON.
pnpm audit:seo must report missing canonicals or descriptions, duplicate titles, orphan pages and
sitemap URLs returning non-200.
ACCEPTANCE: audit:seo clean; Rich Results passes for listing, market, report and organisation; no
gated URL appears in any sitemap; llms.txt served as text/plain; every market page has a unique title
and description and at least three sourced statistics.
```

Prompt 11 — Enquiries, desk, brochures, dashboards

A) Enquiry form used on listing, off-market, market and contact pages: Zod schema shared client and server, Turnstile, honeypot, timing check, rate limiting (5/IP/hour), consent stored with timestamp and IP, UTM captured. Routing to the desk and the listing agent via Resend using the React Email templates in §12.5, sent from our domain with reply-to the enquirer. Member enquiries attach the memberId. Sample listings log but never email. Cal.com embed for desk calls.

B) Brochure PDF in /src/lib/pdf: A4, localised, token-styled, audience-aware (a public brochure omits member-only content), with cover image, logo, price per priceDisclosure, fact table, provenance, 6 gallery images, a static locality map and the disclaimer footer. Exposed at /api/property/[slug]/brochure.pdf and as an admin action.

C) Admin dashboard per §9.2 and the member-facing account dashboard. All queries via /src/lib/db.

ACCEPTANCE: the PDF restyles when tokens change; a public brochure contains no member-only field; dashboards load under 1s with the sample dataset; an enquiry routes, emails and appears in the queue.

Prompt 12 — Sample data and content production

Implement spec §13.10 using the image rules in §13.2, the description grammar in §13.8 and the safety rules in §13.12: 45 properties (30 public, 15 off-market) across the 18 markets with the correct tier weighting and exactly four prime exceptions, all three priceDisclosure modes represented, 8 fictional agencies with 16 agents on monogram avatars, 8 test members across every status with credentials only in .env.local, 12 saved listings, populated MemberActivity, 18 market records with clearly-marked sample statistics, 3 reports, 8 articles, ~30 segment pages.

Deterministic seed, idempotent upsert, --count/--market/--wipe flags, summary table.

Add prnm sample:purge removing every sample record, member, media asset and activity row.

ACCEPTANCE: running twice is idempotent; every filter returns at least three results; each sample member demonstrates their state on login; no sample content is indexable; purge leaves zero orphans.

Prompt 13 — Accessibility and performance passes

A) WCAG 2.2 AA per spec §16, including join, login, the account area and the off-market section. axe-core on 10 routes in CI; commit a manual keyboard-only test script.

B) Performance per spec §14: audit public routes for accidental dynamic rendering and member routes for accidental caching; bundle analysis with dynamic imports for map, lightbox, charts and PDF; explicit sizes on every next/image with exactly one priority image per page; cache headers with a verified edge HIT on public pages; the signed /api/revalidate webhook from Payload afterChange; Lighthouse CI budgets as hard failures.

Report before/after numbers for every budget and list the top five remaining offenders.

ACCEPTANCE: zero axe violations on all 10 routes; Lighthouse accessibility 100 on home, collection and listing; every §14 budget met; no member route ever returns a public cache header.

Prompt 14 — Security, tests, staging, launch, handover

Implement spec §17, §19 and §20.

Security headers with a strict CSP compatible with MapLibre and Plausible; rate limits; Turnstile; staff 2FA; signed webhooks; Sentry with PII scrubbing and no request bodies on member routes; dependency audit in CI. Privacy: granular consent with a stored record, self-service account deletion, one-click admin anonymisation preserving pseudonymised activity, retention jobs per §17.5, and /docs/privacy-procedures.md.

Tests: the full enumerated projection table, the E2E suite in §19 including signed-URL expiry and off-market exposure assertions, visual regression baselines, Plausible events per §18 on the public layer only, and an external uptime monitor.

Deploy to a password-protected Vercel staging domain and document every env var.

Produce LAUNCH-CHECKLIST.md (blocking items: audit:exposure green, no sample content reachable, SAMPLE_DATA_ENABLED=false, legal placeholders replaced, §13.13 content checklist green), README.md (new developer running in 10 minutes), /docs/admin-guide.md, /docs/architecture.md, and a recorded walkthrough script covering: publish a public listing, publish an off-market listing, approve nothing (explain the flag), read the dashboards, add a market page.

ACCEPTANCE: securityheaders.com grade A; CSP clean on every page including the member area; CI fully green; every checklist item ticked or explicitly deferred in DECISIONS.md.

Prompt 15 — TRACK B: agency accounts and roles (optional)

Implement spec §9.4 part one.

- ONE reusable tenant access function in /src/payload/access/tenant.ts applied to every collection.
- Roles agency_admin and agency_agent with the permission model in §9.4: agencies may create and edit their own listings but saving sets moderation = unreviewed; they never publish directly and never see another agency's data.
- Agency onboarding: application at /sell reviewed by an admin, agency record created, set-password email, profile completeness gate before the first submission.
- An agency dashboard scoped to their own listings, views, enquiries and response times.

ACCEPTANCE: Playwright proves agency B receives 403 or empty results on agency A's listings, media and enquiries through both the admin UI and the REST/GraphQL API; an agency user cannot set status to published; an editor can.

Prompt 16 — TRACK B: moderation and bulk import (optional)

Implement spec §9.4 part two.

- Review queue at `/admin/review`: submitted data side by side with automated checks (threshold including `internalValueEur`, ≥ 8 images at $\geq 2000\text{px}$, `description` ≥ 600 characters, no contact details in public text, coordinates plausible for the stated market, EXIF stripped, duplicate fingerprint matches) and a publication preview showing what a visitor and a member would see. Approve, request changes, or reject with a note that reaches the agency.
 - CSV/XLSX bulk import: upload $\rightarrow$ dry-run validation report with row and column level reasons $\rightarrow$ confirm $\rightarrow$ background job $\rightarrow$ summary email with a downloadable error CSV. Idempotent on (agency, reference). Images fetched, validated, optimised and deduplicated. `ImportJob` audit record.
 - Import may set `channel = public` only; off-market listings must be entered manually.
- ACCEPTANCE: importing the sample CSV twice creates no duplicates; a row priced at €4M is rejected with a clear reason; a row attempting `channel = off_market` is rejected; a listing with 5 images is blocked by moderation.

23. Deferred features — exactly how each one switches on

Nothing here is built now. Each is listed with the seam that already exists, so the work is additive.

23.1 Member vetting and approval

Seam: `Member.status` already supports `pending`; `MEMBER_REQUIRE_APPROVAL` already gates the join flow; `/admin/members` already lists accounts. **Work:** an approval queue UI, two email templates, and turning the flag on. Half a day.

23.2 NDA and financial capability confirmation

Seam: `Member.ndaStatus`, `ndaSignedAt`, `capabilityStatus`, `capabilityBand`, `tier` exist and are unused; `/lib/services/` is the integration boundary; the projection table already supports more than two audiences. **Work:** an e-signature integration (Dropbox Sign, DocuSign, Yousign), a capability upload with private storage, an admin review screen, tier promotion, and a fourth column in the projection table. **If this is ever built, a DPIA becomes necessary** and capability documents must be deleted on verification, with only the outcome retained — design it that way from the first line. Roughly two weeks.

23.3 Per-member forensic watermarking

Seam: all member media already flows through `/api/secure/media/[token]` with the member id in scope, and `MemberActivity` already records every access. **Work:** a `sharp` composite for images and a `pdf-lib` overlay for documents carrying the member's name and a token, cached per member-asset pair for 24 h, with the token recorded in the activity row. Two to three days, and it is what convinces the most cautious sellers to place inventory.

23.4 Named per-listing access

Seam: `canSee()` is a single function, and the projection table is data, not code. **Work:** an `AccessGrant` collection, one clause in `canSee`, an admin action, and an expiry job. Two days.

23.5 Agency feeds

Seam: `Property.sourceType` and `ImportJob` exist; the import validation pipeline arrives with Track B. **Work:** per-agency tokens, a documented inbound JSON schema, adapters for common formats, and a cron diff report. Public channel only. About a week.

23.6 Semantic matching and AI search

Seam: `/src/lib/match.ts` is one isolated function; no embedding column exists yet, deliberately, because it is machine-generated and a script can backfill it in one run. **Work:** enable `pgvector`, add the column, backfill, swap the scoring function. Two days.

23.7 CRM, messaging, data rooms, payments

Seam: `/lib/services` is empty and is the only place external integrations may attach; `Enquiry` and `Document` already model the data. All are additive.

24. Appendix

24.1 Reference links

Primary: `christiesrealestate.com` · `sothebysrealty.com` · `jamesedition.com/real_estate` · `knightfrank.com` and `knightfrank.com/wealthreport` · `savills.com/research` · `conciergeauctions.com` · `vladi-private-islands.de/en` · `privateislandsonline.com` · `barnes-international.com` · `engelvoelkers.com` · `luxuryestate.com` · `lionard.com`

Patterns: `airbnb.com` · `compass.com` · `rightmove.co.uk` · `1stdibs.com` · `sothebys.com` · `stripe.com` · `linear.app` · `mercury.com` · `wise.com`

Data sources (§13.6): [agenziaentrate.gov.it \(OMI\)](#) · [istat.it](#) · [landregistry.data.gov.uk](#) · [data.gouv.fr \(DVF\)](#) · [notaires.fr](#) · [ine.es](#) · [mitma.gob.es](#) · [ine.pt](#) · [bankofgreece.gr](#) · [bfs.admin.ch](#) · [fred.stlouisfed.org](#) · [redfin.com/news/data-center](#) · [zillow.com/research/data](#) · [dubailand.gov.ae](#) · [bis.org/statistics/pp.htm](#) · [ec.europa.eu/eurostat](#) · [data.oecd.org](#)

Content sources (§13.2-13.5): [unsplash.com/documentation](#) · [pexels.com/api](#) · [pixabay.com/api/docs](#) · [api.openverse.org](#) · [geonames.org](#) · [naturalearthdata.com](#) · [overpass-turbo.eu](#) · [nominatim.org](#) · [fonts.google.com](#) · [gwfh.mranftl.com](#) · [fontsource.org](#) · [lucide.dev](#) · [realfavicongenerator.net](#) · [api.frankfurter.app](#)

Docs: [nextjs.org/docs](#) · [payloadcms.com/docs](#) · [postgis.net/documentation](#) · [typesense.org/docs](#) · [maplibre.org/maplibre-gl-js/docs](#) · [docs.maptiler.com](#) · [docs.protomaps.com](#) · [developers.cloudflare.com/r2](#) · [developers.cloudflare.com/turnstile](#) · [upstash.com/docs](#) · [next-intl.dev](#) · [react-pdf.org](#) · [react.email](#) · [cal.com/docs](#) · [schema.org/RealEstateListing](#) · [schema.org/Dataset](#) · [web.dev/vitals](#) · [w3.org/WAI/WCAG22/quickref](#) · [llmstxt.org](#)

24.2 Glossary

Channel whether a listing is public or off-market · **Projection** the allowlist of fields serialised for a given audience · **Prime entry price** the threshold at which a market's top 5 % begins · **SPV** special-purpose vehicle holding a property · **Share transfer** selling the company that owns the asset rather than the asset · **Requirements profile** a member's stated search brief · **asOfDate** the date a published statistic refers to · **ISR** incremental static regeneration.

24.3 One-page summary for the client

Question	Answer
What is it?	A fast, editorially serious portal for property from €20 M, fully public, with an off-market section that opens with a free account
What does a visitor have to do to see the off-market properties?	Enter an email address and click a confirmation link. Nothing else
Why will it rank?	A complete public collection plus a market-intelligence library with sourced, dated figures — the kind of content this audience searches for and assistants cite
Why will people register?	The off-market inventory and the report PDFs. Registration is the conversion event, and it produces an owned audience instead of rented traffic
Why will brokers place inventory here?	Control over publication per listing, and a demand report showing how many members viewed their property
What is deliberately not built yet?	NDA, proof of funds, vetting, watermarking, agency accounts — each with a documented seam in §23, none requiring a rewrite
Time to live	~8–9 weeks for Track A with one senior developer using Cursor + Claude; ~6 weeks with two
Client effort	Domain, logo approval, legal entity details, legal review of the policy placeholders, staging sign-off. No photo or copy curation

End of specification v2.0. Every deferred feature in §23 attaches to a seam that already exists in this build.